

Cover Page

Acknowledgement

Abstract

Contents

1. Introduction	1
1.1. Purpose.....	1
1.2. Project Scope	1
1.3. Document Convention.....	1
1.4. Additional Information.....	1
2. Background and Literature Review	1
3. Overall Description.....	3
3.1. Product perspective	3
3.2. Product Feature	3
3.3. User Classes and Characteristics.....	3
3.4. Operating Environment	4
3.5. Design and Implementation Constraints	4
3.6. Assumptions and Constraints	4
4. External Interface Requirements.....	6
4.1. User Interfaces	6
4.2. Hardware Interfaces.....	6
4.2. Software Interfaces	7
4.3. Interaction between Hardware and Software:	7
5. Other Requirements	9
6. Non Functional Requirements.....	16
6.1. Performance Requirements.....	16
6.2. Safety Requirements	16
6.3. Security requirements.....	16
6.4. Software Quality Attributes	17
6.5. Gantt chart.....	17
7. Conclusion	18
8. Recommendations.....	18
9. Future Scope:	18

Figure 1 UML Class Diagram for E-learning Platform	9
Figure 2 Use Case Diagram for E- learning platform	10
Figure 3 ER diagram for E-learning Platform	11
Figure 4 Context level DFD.....	12
Figure 5 Level 1 DFD	12
Figure 6 Level 2 DFD	13
Figure 7 Activity Diagram of E- learning Platform.....	14
Figure 8 Sequence Diagram.....	15
Figure 9 Gantt Chart	17

1. Introduction

In this Era of Science of Technology there are lots of developed online learning platform offering different courses. As with the existence of E-learning students/ learners can access to study materials and student can have better understanding of content. Web-based studying has been considered a passive activity until recently. An online learning platform is a web-based system that provides educational content and resources to learners. Online learning platforms can offer various types of courses with notes, lab reports and Important questions. Some Examples of online learning platforms are Udemy, Coursera, ABC Mouse etc. Online learning platforms are a great way to explore new topics and enhance knowledge. E-learning platforms suits the needs of goals of learner

1.1. Purpose

A E-Learning Platform is a software designed to manage provide students with different features and benefits such as Immediacy, Low cost, self-paced, comfortability etc.

1.2. Project Scope

These days Students like to study in their own pace and time. The scope of our project is to design, develop and test the e – learning platform for subject areas of BSc. CSIT and provide time flexibility along with user-accessibility.

1.3. Document Convention

The document is prepared using Microsoft Word 2016 and has used the font type 'Times New Roman'. The fixed font size that has been used to type this document is 12pt and for headings 16pt with 1.5 lines pacing. It has used the bold property to set the headings of the document. All pages except the cover page, abstract, acknowledgement, table of content and list of figures are numbered, the numbers appear on the bottom of the page. Every image and data table are numbered and referred to the in the main text.

1.4. Additional Information

E-learning System can be used by entering respective username and password by the users. It is accessible either by an administrator or professor. Only the respective person can add data in the

database. The data can be retrieved easily. The interface is very user-friendly. The data are well protected and data processing is very fast, accurate and relevant.

2. Background and Literature Review

2.1. Background study

In the contemporary educational landscape, E-learning platform emerges as a transformative force in the field of e-learning, particularly tailored to cater to the diverse needs of students in computer science and information technology. As technological advancements continue to shape how knowledge is acquired and disseminated, E-learning stands at the forefront, offering a comprehensive array of resources and tools aimed at facilitating effective learning experiences.

The rapid advancement of digital technologies has led to a paradigm shift in educational methodologies with the emergence of e-learning. E-learning has grown in popularity as a result of the widespread availability of mobile devices and internet connectivity, giving students the freedom to interact with instructional materials whenever and wherever they choose. Computer science and information technology have become foundational disciplines with broad implications across multiple sectors in an era of rapid technological innovation. Proficiency in fields such as software development and cybersecurity is crucial for effectively navigating the intricacies of the digital era. E-learning is dedicated to giving students the information, tools, and resources they need to succeed in a world that is becoming more and more digital because it recognizes the importance of computer science and information technology education.

2.2. Literature Review

E-learning platforms have the potential to enhance student engagement and improve learning outcomes (Bonk & Graham, 2006). By providing learners with control over the pace, place, and path of their learning, e-learning platforms empower individuals to take ownership of their education and pursue personalized learning trajectories (Bates, 2019). In the context of computer science and information technology education, e-learning platforms have emerged as invaluable resources for students seeking to acquire technical skills and competencies. With the rapid pace of technological innovation, there is a growing demand for professionals proficient in areas such as programming, data analysis, and cybersecurity (Hunt, 2019). E-learning platforms specializing in computer science and information technology education play a crucial role in addressing this demand by providing learners with access to high-quality educational content and practical learning experiences (Chen & Hwang, 2014).

2.3. Existing System

The current e-learning environment consists of a wide range of platforms that serve different target audiences and educational domains. In the field of teaching computer science and information technology, a number of well-known platforms have made a name for themselves. Leading universities and business professionals teach a variety of computer science-related courses and specialization programs on platforms like Coursera, edX, and Udacity. Additionally, coding-focused websites such as LeetCode, Codecademy, and freeCodeCamp concentrate especially on the development of programming skills; they provide learners with interactive coding challenges, tutorials, and exercises to help them improve their problem-solving and coding proficiency. The importance of ongoing research and development in the field of e-learning platforms is highlighted by difficulties including making learning accessible to students from a variety of backgrounds, keeping students' attention for extended periods of time, and meeting student requirements.

3. Overall Description

A E-Learning platform is a webpage designed to provide the study materials required to Bsc.CSIT students. E-learning simply means online learning platform. It is an online site that keeps track of students and stores study materials for student.

3.1. Product perspective

An online learning platform emphasizes and presents the learner's perspective. It provides a safe learning environment where students can take online courses. An online learning platform is a web-based system that provides educational content and resources to learners.

3.2. Product Feature

Admin Module:

- Add study material
- Delete study material
- Update study material (Make changes)
- Host Online Exams

Student Module:

- Access study materials
- Search for study materials
- Display notes and Important Questions
- Download contents
- Take Exams

3.3. User Classes and Characteristics

The system will be used in the College and Home. The administrators, Students will be the main users. Given the condition that not all the users are computer-literate. The system is user- friendly.

3.4. Operating Environment

The system is also designed to be user-friendly. The Webpage will operate, in every browser with internet connection which it must peacefully coexist

3.5. Design and Implementation Constraints

- Develop web based tool for managing learners' courses.
- Develop a feature which will be used to make sure that all the levels of learning are covered.
- Design and integrate a feature which will enable learners to answer questions based on their learning abilities for evaluations
- Design various database structures
- Perform documentation of design

3.6. Assumptions and Constraints

Assumptions:

- The students must have little knowledge of working and learning online.
- The college and student have reliable internet connectivity to access the webpage.
- Student must have schedule online time commitments.
- learners are motivated and self-disciplined enough to engage with the materials and complete coursework independently.
- Content and technology need to be regularly updated to keep pace with changes in the subject matter, technology, and learning needs.
- The learner has access to a suitable environment, free from major distractions, where they can concentrate on learning.

Constraints:

- Financial and human resources, including budget limitations, and the availability of instructional designers, instructors, and support staff.
- The time, skills, and resources required to create high-quality e-learning content, including videos, quizzes, and interactive modules.
- Assessing and evaluating learner progress and the effectiveness of the e-learning platform, including limitations in assessment tools and methods.

- Keeping learners engaged and motivated throughout the learning process, particularly in self-paced e-learning courses.
- The availability and reliability of the internet and related infrastructure, especially in areas with limited connectivity.
- Scaling the platform to accommodate a growing number of learners, which may require additional investment in infrastructure and resources.

4. External Interface Requirements

4.1. User Interfaces

Gives a description of the logical features of each user-software product interface. Sample screen shots, any GUI guidelines or product family style manuals that must be adhered to, restrictions on screen arrangement, and standard buttons and features (like help) that will appear on all screens, keyboard shortcuts, standards for the display of error messages, and so forth. Specify the elements of software that require an interface. Specifications of the UI design ought to be recorded in an independent user interface specification.

4.2. Hardware Interfaces

Gives a description of the logical and physical properties of each interface that the software product has with the system's hardware. This could include the kinds of devices that are supported, the type of data that is being transferred, how the software and hardware interact, and protocols for communication to be employed.

- **Computers/Devices:**

Students and instructors use various devices such as desktops, laptops, tablets, and smartphones to access the e-learning platform.

- **Input Devices:**

Keyboards, mice, touchscreens, and other input devices are essential for users to interact with the e-learning platform.

- **Output Devices:**

Monitors, speakers, and printers are used for displaying content, listening to audio, and printing materials if necessary.

- **Networking Equipment:**

Routers, switches, and internet connectivity are crucial for accessing online courses, multimedia content, and collaborative tools.

4.2. Software Interfaces

Gives a brief explanation of the relationships this product has with other particular software components (name and version), such as operating systems, databases, tools, libraries, and integrated commercial components. List the messages or data items entering and leaving the system along with their respective purposes. Describe the type of communications and the services that are required. Consult the documentation for a thorough explanation of the application programming interface protocols. Determine which data will be shared by different software components. Indicate this as an implementation constraint if the data sharing mechanism needs to be implemented in a particular way (using a global data area in an operating system that allows multitasking, for example).

- **Content Management System (CMS):**

Manages the creation, storage, and organization of learning materials, including text, multimedia, and assessments.

- **Security and Authentication Software:**

Ensures the security and integrity of the e-learning platform by implementing user authentication, data encryption, and access controls.

- **Analytics and Reporting Tools:**

Software that provides insights into student performance, engagement, and learning patterns. This information helps instructors and administrators make data-driven decisions.

4.3. Interaction between Hardware and Software:

- **User Interface (UI):**

The interface through which users interact with the software, designed to be user-friendly across various devices and screen sizes.

- **Data Transmission:**

The seamless exchange of data between the e-learning platform and users' devices, facilitated by reliable networking infrastructure.

- **Compatibility:**

Ensuring that the software is compatible with different operating systems, browsers, and devices to provide a consistent experience for all users.

- **Scalability:**

The ability of the hardware and software to scale and accommodate a growing number of users and increasing data volumes.

- **Updates and Maintenance:**

Regular updates to both hardware and software components to enhance functionality, security, and performance.

The effective integration of hardware and software components is essential for creating a robust and user-friendly e-learning environment that supports diverse learning needs.

5. Other Requirements

UML Class Diagram for E-learning Platform:

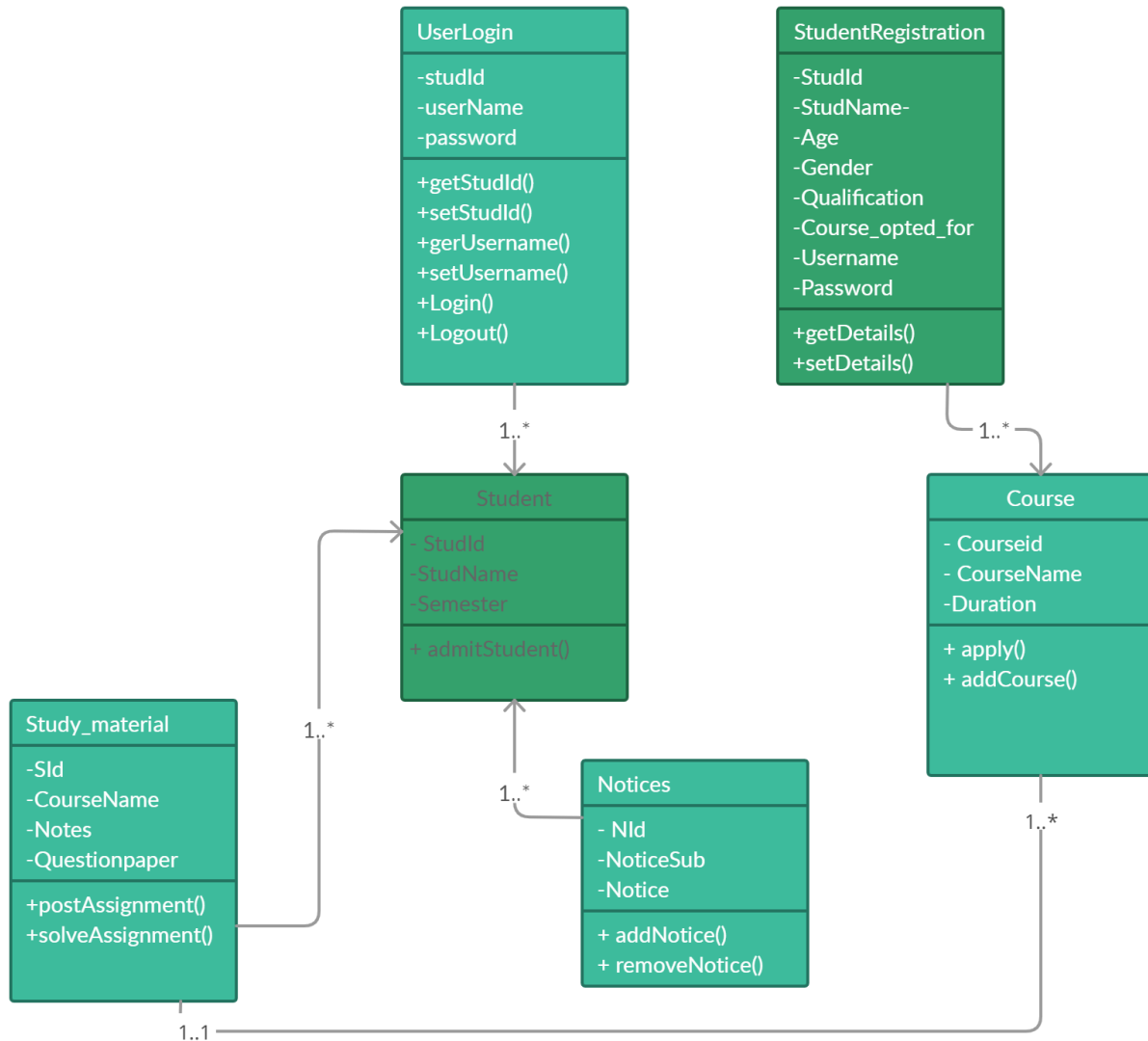


Figure 1 UML Class Diagram for E-learning Platform

Use Case Diagram

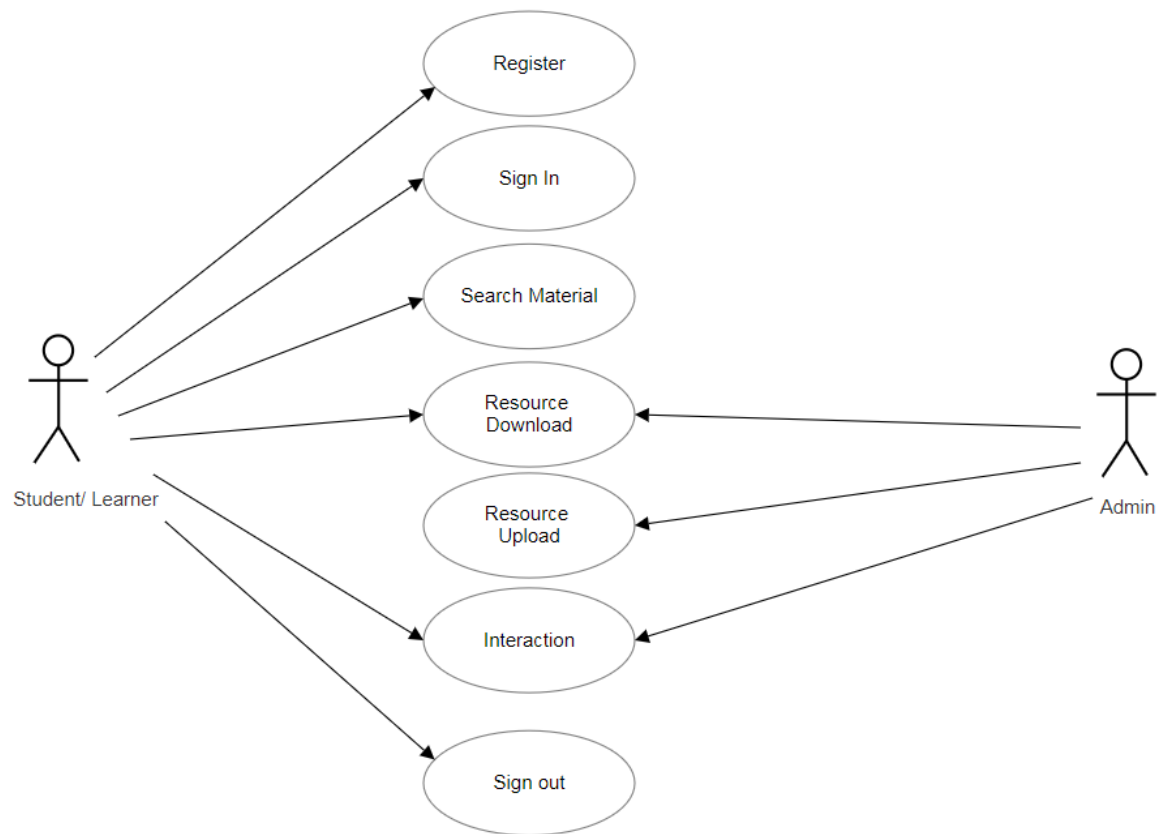


Figure 2 Use Case Diagram for E- learning platform

ER- Diagram:

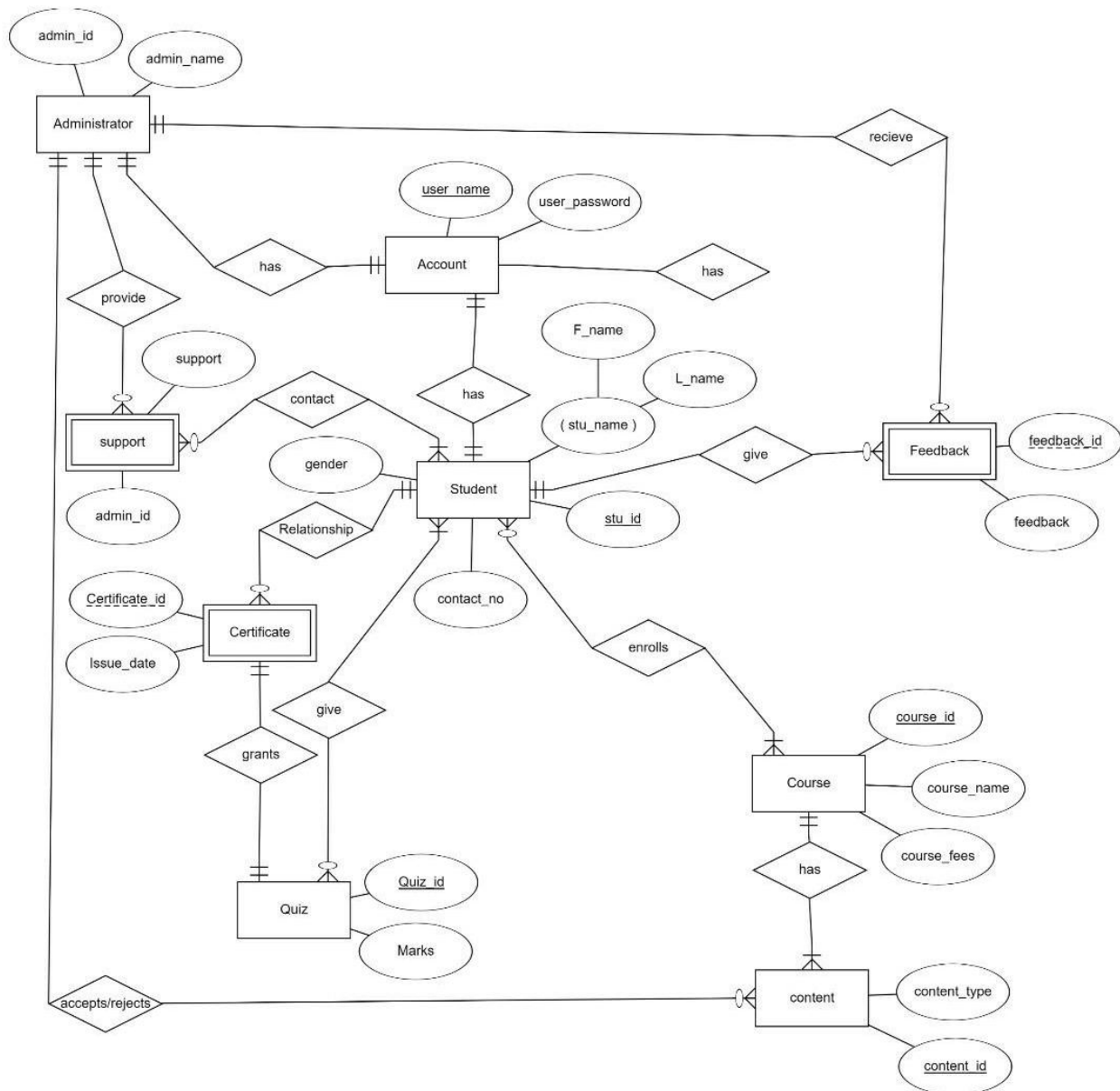


Figure 3 ER diagram for E-learning Platform

Context Level DFD:

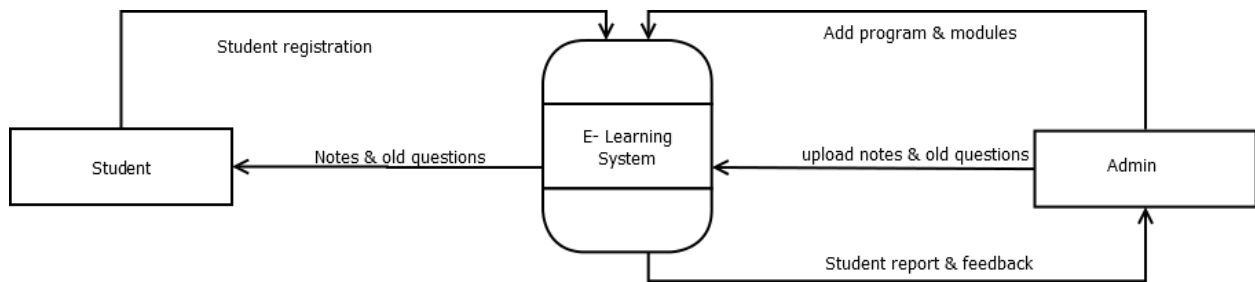


Figure 4 Context level DFD

Level 1 DFD:

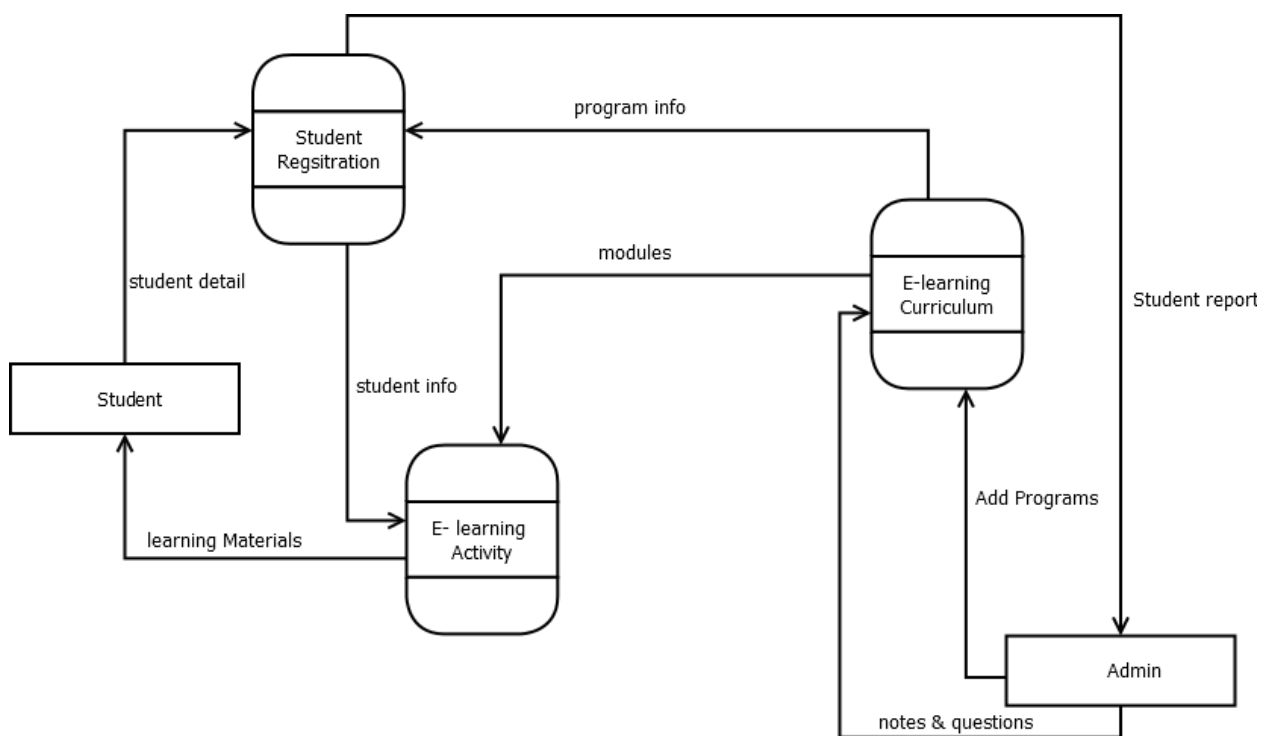


Figure 5 Level 1 DFD

Level 2 DFD:

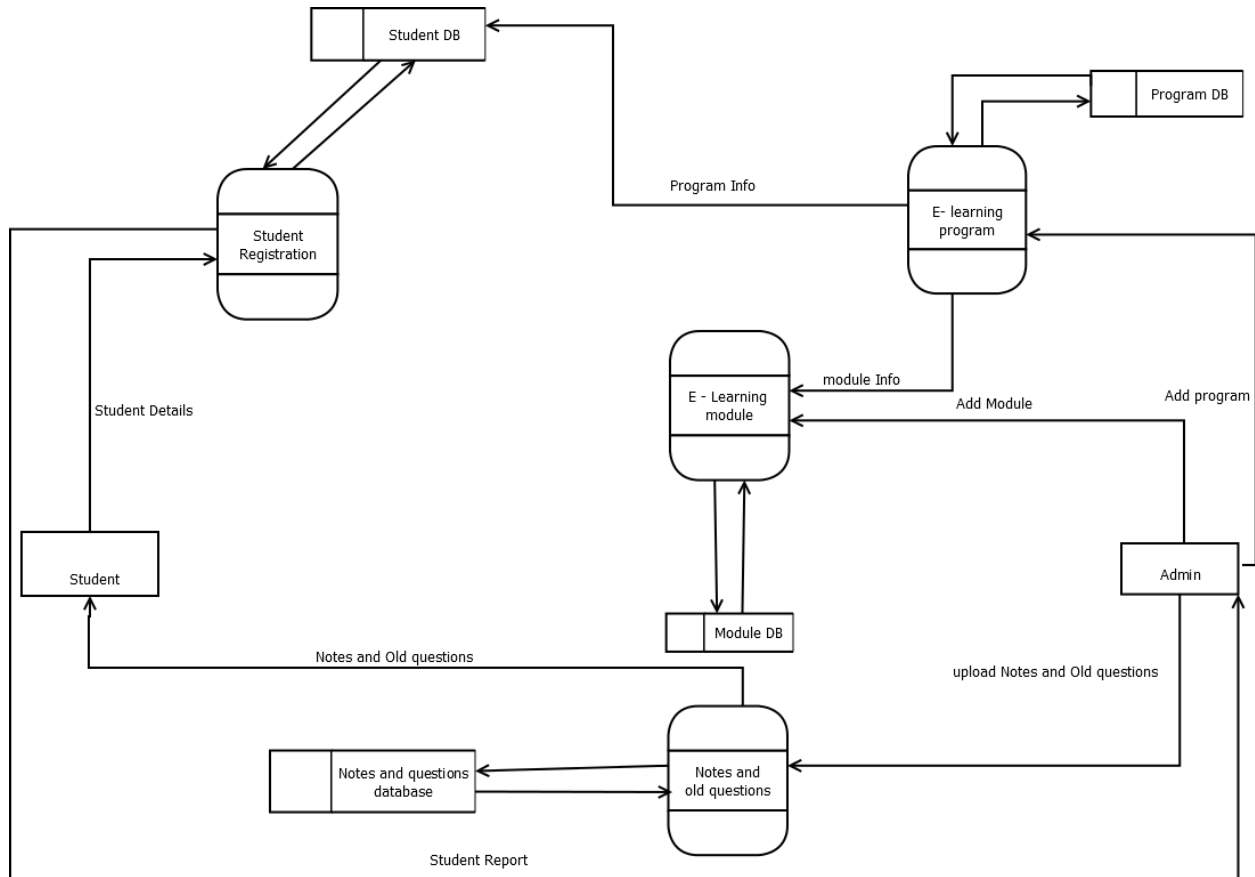


Figure 6 Level 2 DFD

Activity Diagram:

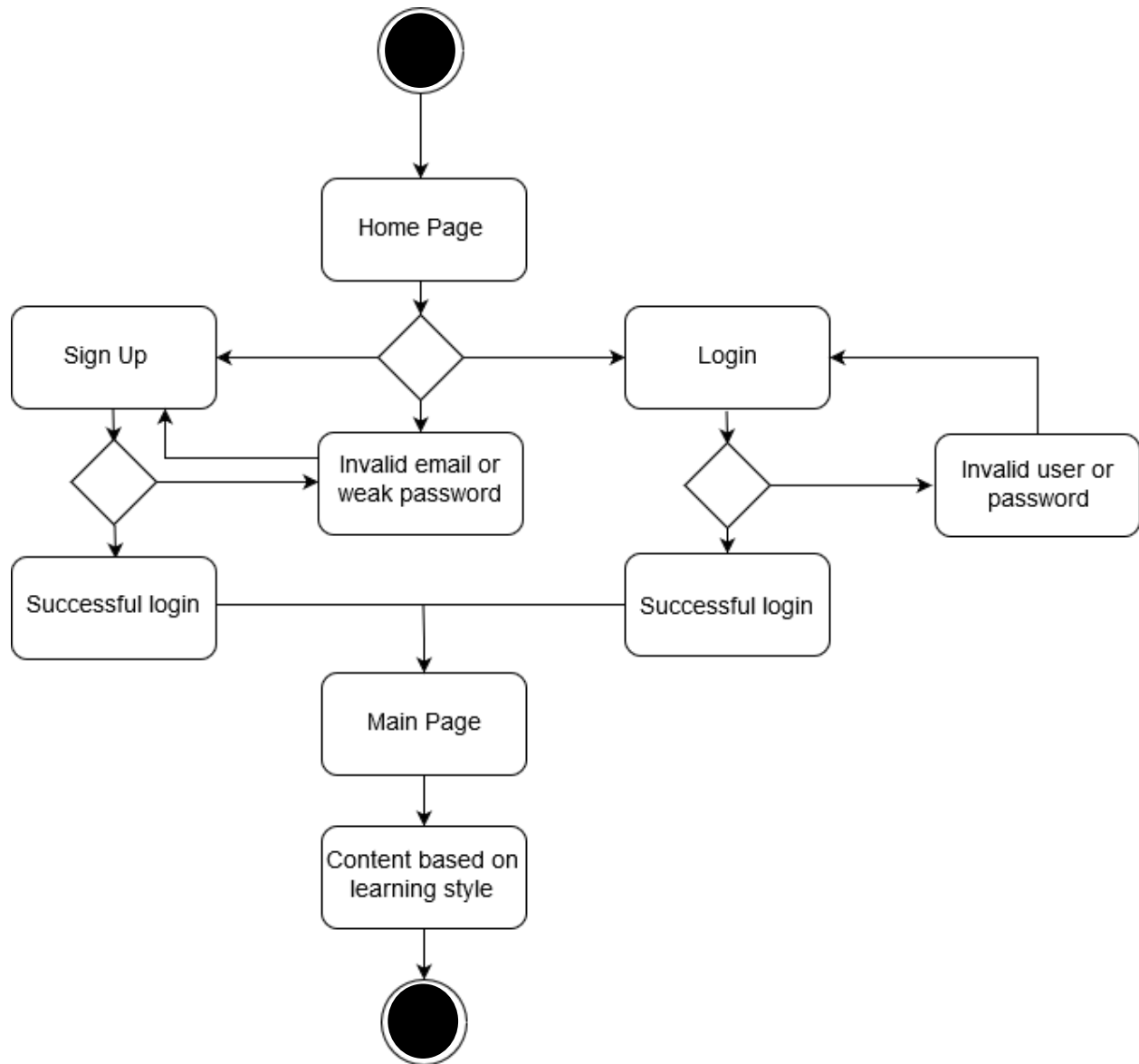


Figure 7 Activity Diagram of E- learning Platform

Sequence Diagram

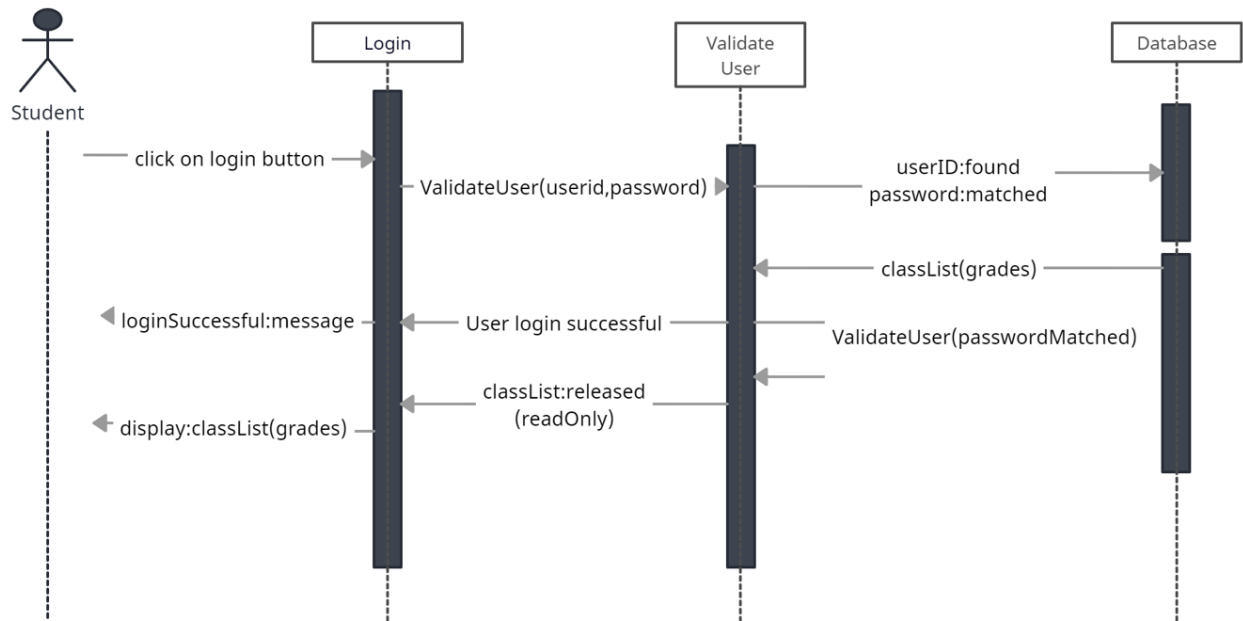


Figure 8 Sequence Diagram

6. Non Functional Requirements

6.1. Performance Requirements

- Response Time-Specify that web pages, course modules, and multimedia content must load within a defined maximum time (e.g., 2 seconds) to ensure a seamless learning experience.
- Availability: Specify the level of availability (e.g., 99.9%) the platform must achieve to minimize downtime for maintenance.
- Security Performance-Ensure that encryption and authentication processes do not significantly impact response times during secure transactions.
- Browser Compatibility- Ensure consistent performance across various web browsers and versions to support a wide range of users.
- Resource Utilization –Optimize server resource utilization to ensure efficient handling of user requests and content delivery.

6.2. Safety Requirements

Safety requirements in the context of an e-learning platform typically focus on ensuring the security and well-being of users, protecting their data and privacy, and complying with relevant regulations and best practices. Some safety requirements are:

- Data Security using encryption and encryption technique
- User privacy
- Content security
- Authenticate user
- Phishing prevention
- Secure communication
- Vulnerability management
- User Awareness

6.3. Security requirements

Security requirements for an e-learning platform encompass measures to safeguard user data, privacy, and the integrity of the system. These include robust encryption for data transmission and

storage, secure user authentication, protection against phishing and cyber threats, prompt incident response procedures, compliance with data protection and accessibility regulations, as well as measures to prevent intellectual property violations and ensure safe online behavior. It's essential to conduct regular security audits and provide user education on security best practices to maintain a safe and trusted learning environment.

6.4. Software Quality Attributes

- **Availability:** The system shall be available all the time.
- **Correctness:** A bug free software which fulfill the correct need/requirements of the client.
- **Maintainability:** The ability to maintain, modify information and update fix problems of the system 21
- **Usability:** software can be used again and again without distortion.
- **Accessibility:** Administrator and many other users can access the system but the access level is controlled for each user according to their work scope.
- **Accuracy:** The reliability on the information/output. Can depend/be sure of the outcome.
- **Stability:** The system outcome/output won't change time to time. Same output will be given always for a given output.

7. Gantt chart

	1 st week	2 nd week	3 rd week	4 th week	5 th week	6 th week	7 th week
Planning							
Analysis							
Design							
Coding							
Testing							
Implementation							
Documentation							

Figure 9 Gantt Chart

8. Conclusion

In conclusion, the Software Development Life Cycle (SDLC) document for the E-learning Platform provides a comprehensive framework for the development, deployment, and maintenance of an effective online learning solution. The systematic approach outlined in the SDLC ensures that all stages of development are well-planned, executed, and monitored. The successful implementation of the SDLC is crucial for the overall success of the E-learning Platform, ensuring that it meets the needs of users, adheres to quality standards, and remains adaptable to future changes.

9. Recommendations

Regular Testing and Quality Assurance: Conduct thorough testing at each stage of the SDLC to identify and rectify any bugs or issues. Implement automated testing procedures and ensure rigorous quality assurance to deliver a stable and reliable E-learning Platform.

Scalability and Performance Optimization: Plan for future scalability by designing the platform to handle a growing user base. Regularly optimize performance to ensure fast loading times and a seamless learning experience, especially during peak usage periods.

Security Measures: Continuously monitor and enhance security protocols to safeguard sensitive user data and prevent unauthorized access. Regularly update security measures to stay ahead of potential threats and vulnerabilities.

Adaptability to Technological Changes: Stay abreast of emerging technologies and trends in e-learning. Ensure the platform remains adaptable to incorporate new features, integrations, and advancements in educational technology.

10. Future Scope:

Looking ahead, the E-learning Platform has significant potential for evolution. Integration of artificial intelligence and machine learning can personalize learning experiences, while exploring block chain may enhance credential verification. Mobile learning optimizations, collaborative features, and advanced analytics are avenues for improvement.